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PACKAGES WITH ENHANCED HEAT DISSIPATION AND PARASITIC INDUCTANCE MITIGATION

Abstract

In examples, a semiconductor package comprises a semiconductor die having a device side including circuitry and a non-device side opposite the device side; a metal pillar coupled to the device side and extending away from the semiconductor die; and a routing structure coupled to the metal pillar. The routing structure includes first and second metal layers, the first metal layer positioned closer to the device side than the second metal layer and coupled to the second metal layer by way of a via; and a dielectric covering the first and second metal layers and the via. The package includes a die pad coupled to the non-device side of the semiconductor die by way of a thermally conductive material; a die pad extension member coupled to an opposite surface of the die pad than the semiconductor die; and a conductive terminal having a first horizontal member and a second horizontal member coupled to and vertically offset from the first horizontal member, with the second horizontal member soldered to the first metal layer of the routing structure. The package comprises a mold compound covering the semiconductor die, the metal pillar, the routing structure, the die pad, and the conductive terminal, with the first horizontal member and the die pad exposed to an exterior surface of the mold compound.

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Background/Summary

BACKGROUND

[0001] Semiconductor wafers are circular pieces of semiconductor material, such as silicon, that are used to manufacture semiconductor chips. Generally, complex manufacturing processes are used to form numerous integrated circuits on a single wafer. The formation of such circuits on a wafer is called fabrication. After wafer fabrication, the wafer is cut into multiple pieces, called semiconductor dies, with each die containing one of the circuits. The cutting, or sawing, of the wafer into individual dies is called singulation. An individual die is then coupled to a die pad and to conductive terminals, sometimes called “leads.” The resulting structure is subsequently covered with a mold compound to produce a package.

SUMMARY

[0002] In examples, a semiconductor package comprises a semiconductor die having a device side including circuitry and a non-device side opposite the device side; a metal pillar coupled to the device side and extending away from the semiconductor die; and a routing structure coupled to the metal pillar. The routing structure includes first and second metal layers, the first metal layer positioned closer to the device side than the second metal layer and coupled to the second metal layer by way of a via; and a dielectric covering the first and second metal layers and the via. The package includes a die pad coupled to the non-device side of the semiconductor die by way of a thermally conductive material; a die pad extension member coupled to an opposite surface of the die pad than the semiconductor die; and a conductive terminal having a first horizontal member and a second horizontal member coupled to and vertically offset from the first horizontal member, with the second horizontal member soldered to the first metal layer of the routing structure. The package comprises a mold compound covering the semiconductor die, the metal pillar, the routing structure, the die pad, and the conductive terminal, with the first horizontal member and the die pad exposed to an exterior surface of the mold compound.

[0003] In examples, a method for manufacturing a package comprises coupling a metal pillar extending from a device side of a semiconductor die to a routing structure, with the device side having circuitry formed therein, the routing structure comprising first and second metal layers, the first metal layer positioned closer to the device side than the second metal layer and coupled to the second metal layer by way of a via, and a dielectric covering the first and second metal layers and the via. The method comprises coupling a non-device side of the semiconductor die opposing the device side of the semiconductor die to a die pad by a thermally conductive material; coupling a conductive terminal to the routing structure; and covering the semiconductor die, the die pad, the metal pillar, the routing structure, and the conductive terminal with a mold compound, with the conductive terminal exposed to an exterior surface of the mold compound.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a flow diagram of a method for manufacturing a semiconductor package, in accordance with various examples.

[0005] FIGS. **2**A**1**, **2**A**2**, **2**A**3**, **2**A**4**, and **2**A**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and a semiconductor die, in accordance with various examples.
[0006] FIGS. **2**B**1**, **2**B**2**, **2**B**3**, **2**B**4**, and **2**B**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a die pad, die pad extension member, and conductive terminals, in accordance with various examples.
[0007] FIGS. **2**C**1**, **2**C**2**, **2**C**3**, **2**C**4**, and **2**C**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples.
[0008] FIGS. **2**D**1**, **2**D**2**, **2**D**3**, **2**D**4**, and **2**D**5** are profile, alternative profile, top-down, bottom-up, and perspective views of a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples.
[0009] FIGS. **2**E**1** and **2**E**2** are profile and perspective views of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples.
[0010] FIG. **2**F is a profile, cross-sectional view of a routing structure in accordance with various examples.
[0011] FIGS. **3**A**1**, **3**A**2**, **3**A**3**, **3**A**4**, and **3**A**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and a semiconductor die, in accordance with various examples.
[0012] FIGS. **3**B**1**, **3**B**2**, **3**B**3**, **3**B**4**, and **3**B**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a die pad, die pad extension member, and conductive terminals, in accordance with various examples.
[0013] FIGS. **3**C**1**, **3**C**2**, **3**C**3**, **3**C**4**, and **3**C**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples.
[0014] FIGS. **3**D**1**, **3**D**2**, **3**D**3**, **3**D**4**, and **3**D**5** are profile, alternative profile, top-down, bottom-up, and perspective views of a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples.
[0015] FIGS. **3**E**1** and **3**E**2** are profile and perspective views of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples.
[0016] FIG. **3**F is a profile, cross-sectional view of a routing structure in accordance with various examples.
[0017] FIG. **4** is a block diagram of an electronic device comprising a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples.

DETAILED DESCRIPTION

[0018] Heat dissipation is a recurring technical challenge in the semiconductor packaging industry. Different solutions have been employed to varying effect. In some cases, a “flip-chip” semiconductor die is oriented such that a device side in which circuitry is formed is oriented downward, facing a die pad or other part of a lead frame, and non-bond wire structures such as pillars and bumps are used to couple the device side to the die pad or lead frame. A metal lid is coupled to the back side of the semiconductor die (also referred to as the non-device side of the semiconductor die), and the metal lid is exposed to an exterior surface of the package. In this way, heat is dissipated through the non-device side of the die, the metal lid, and out of the package. While some heat may be dissipated in this manner, the package is coupled to a printed circuit board (PCB), and heat dissipation through the PCB is suboptimal because of the limited contact area between the device side of the semiconductor die and the die pad (i.e., through pillars or bumps). Further, even the heat dissipated through the metal lid is suboptimal because the thermal pad positioned between the metal lid and the non-device side of the semiconductor die is relatively small, thus restricting the amount of heat that can be dissipated through the metal lid. In other cases, the semiconductor die does not have a “flip-chip” configuration, meaning that the non-device side of the semiconductor die is facing downward and is coupled to the die pad, while the device side of the semiconductor die is facing upward and is coupled to conductive terminals (or “leads”) through bond wires. While thermal dissipation through the die pad and the PCB to which the package is coupled may be improved relative to other solutions, the bond wires introduce parasitic inductance into the package. Parasitic inductance in a package can present numerous technical challenges, especially at high frequency of operation, thereby effectively limiting the frequencies of operation that can be achieved. Examples of such technical challenges associated with parasitic inductance can include problems with signal integrity, power supply noise, electromagnetic interference (EMI), switching losses, resonance effects, and heat dissipation challenges, depending on the application.

[0019] This disclosure describes various examples of a package that provides enhanced heat dissipation and parasitic inductance mitigation, thereby resolving the numerous technical challenges described above. In examples, a package includes a semiconductor die having a device side including circuitry and a non-device side opposite the device side. The package includes a metal pillar coupled to the device side and extending away from the semiconductor die. The package includes a routing structure coupled to the metal pillar, with the routing structure including first and second metal layers. The first metal layer is positioned closer to the device side than the second metal layer and is coupled to the second metal layer by way of a via. The routing structure also includes a dielectric covering the first and second metal layers and the via. The package includes a thermally conductive die pad coupled to the non-device side of the semiconductor die by way of a thermally conductive material. The package also includes a conductive terminal having a first horizontal member and a second horizontal member coupled to and vertically offset from the first horizontal member, the second horizontal member soldered to the first metal layer of the routing structure. Further, the package includes a mold compound covering the semiconductor die, the metal pillar, the routing structure, the thermally conductive die pad, and the conductive terminal, with the first horizontal member and the die pad exposed to an exterior surface of the mold compound.

[0020] FIG. **1** is a flow diagram of a method **100** for manufacturing a semiconductor package, in accordance with various examples. The method **100** includes coupling a metal pillar extending from a device side of a semiconductor die to a routing structure (**102**). The device side has circuitry formed therein (**102**). The routing structure comprises first and second metal layers (**102**). The first metal layer is positioned closer to the device side than the second metal layer and is coupled to the second metal layer by way of a via (**102**). A dielectric covers the first and second metal layers and the via (**102**).

[0021] FIGS. **2**A**1**, **2**A**2**, **2**A**3**, **2**A**4**, and **2**A**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of an example routing structure **200** and an example semiconductor die **202** coupled to the routing structure **200**, in accordance with various examples. In examples, the semiconductor die **202** is a gallium nitride (GaN) die, although in other examples, other types of semiconductor materials, such as silicon (Si) or silicon carbide (SiC), may be useful. Further, the circuitry formed in the semiconductor die **202** may include radio frequency (RF) circuitry configured to perform one or more RF operations in an RF application, although the scope of this disclosure is not limited to semiconductor dies that perform RF operations. The principles described herein may be extended to any suitable type of application. The semiconductor die **202** includes a device side **203** in which the circuitry of the semiconductor die **202** is formed, and the semiconductor die **202** also includes a non-device side **205** opposing the device side **203**.

[0022] The routing structure **200** is a multi-layered routing device. The routing structure includes multiple metal layers **204** that are separated from each other by dielectric layers **206**. Metal layers

204 and dielectric layers **206** may be co-located in the same layer, or horizontal plane, of the routing structure **200**. Each of the metal layers **204** may be coupled to another one or more of the metal layers **204** by way of one or more metal vias, thus forming a network of metal layers to carry electrical signals.

[0023] The routing structure **200** is not a printed circuit board (PCB). Unlike a PCB, which typically has metal layers that are separated by air, the dielectric layers **206** of the routing structure **200** are solid and tangible (i.e., not air), such as AJINOMOTO® build-up film (ABF). Furthermore, unlike a PCB, the routing structure **200** is located inside a semiconductor package. PCBs, by contrast, are coupled to and external to semiconductor packages.

[0024] Accordingly, a routing structure is expressly defined herein as a non-PCB structure having a network of two or more metal layers and vias connecting the metal layers, and a solid, tangible, non-gaseous dielectric in contact with the metal layers.

[0025] Referring briefly to FIG. **2**F, which provides a profile, cross-sectional view of the routing structure **200**, the routing structure **200** includes metal layers **204** and dielectric layers **206**, as described above. Vias **212** couple metal layers **204** to each other, thereby forming the above-described network of metal layers **204**. The specific spatial patterns of the metal layers **204** may vary depending on design needs and the particular application in which the routing structure **200** is being deployed. The dielectric layers **206** may comprise a solid, tangible dielectric, such as, for example, ABF, but the dielectric layers **206** do not include air or any other gaseous dielectric. The metal layers **204** include metal members **204***a*, **204***b*, and **204***c*, each of which may couple to a different one of metal pillars **208**.

[0026] The routing structure **200** may be formed by using an iterative process in which a layer of metal is plated, a dielectric layer is applied, and a grinding process is performed to reduce the thickness of the dielectric layer and/or the metal layer and to achieve a top surface of the dielectric layer through which areas of the metal layer are exposed. The process then begins again, until a routing structure with the desired architecture is obtained.

[0027] Metal pillars **208** extend from the device side **203** of the semiconductor die **202**. The metal pillars **208** are coupled to the circuitry formed in the device side **203** and extend away from the device side **203**, toward the routing structure **200**. The metal pillars **208** are coupled to the routing structure **200**, and more specifically, to the network of metal layers **204** (e.g., metal members **204***a*, **204***b*, and **204***c*, and possibly other metal members not visible in the cross-sectional view of FIG. **2**F), by any suitable, conductive adhesive, such as solder bumps **210**.

[0028] Referring again to FIG. **1**, the method **100** includes coupling a non-device side of the semiconductor die opposing the device side of the semiconductor die to a die pad by a thermally conductive material (**104**).

[0029] FIGS. **2**B**1**, **2**B**2**, **2**B**3**, **2**B**4**, and **2**B**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a die pad, die pad extension member, and conductive terminals, in accordance with various examples. In particular, FIGS. **2**B**1**-**2**B**5** show a die pad **220** having a surface **222** and a surface **224** opposite the surface **222**, a die pad extension member **226** coupled to the surface **224** and extending away from the surface **224**, and conductive terminals **228**. At least one of the conductive terminals **228** includes a first horizontal member **230**, a second horizontal member **232**, and a connecting member **234** coupling the first and second horizontal members **230**, **232** to each other. The first and second horizontal members **230**, **232** are vertically offset from each other, meaning that an axial center **236** of the first horizontal member **230** and an axial center **238** of the second horizontal member **232** are vertically distanced from each other. Furthermore, the second horizontal member **232** is more proximal to the die pad **220** than is the first horizontal member **230**, and the first horizontal member **230** is more distal to the die pad **220** than is the second horizontal member **232**.

[0030] The connecting member **234** extends from the first horizontal member **230** to the second horizontal member **232** at an angle relative to the horizontal plane in which the axial center **236** lies. In examples, this angle ranges between **30** degrees and **60** degrees, with a larger (i.e., steeper) angle being disadvantageous because it unnecessarily increases the clearance between the routing structure **200** and the die pad **220**, in addition to requiring a thicker semiconductor die **202** and/or longer metal pillars **208**, and with a smaller (i.e., shallower) angle being disadvantageous because it may leave inadequate clearance between the routing structure **200** and the die pad **220** for the semiconductor die **202** and the metal pillars **208**.

[0031] As described below, upon application of a mold compound, the die pad extension member **226** is exposed to an exterior surface (i.e., bottom surface) of the mold compound. This exposure facilitates heat dissipation away from the semiconductor die **202** and out of the semiconductor package. Thus, the thickness of the die pad extension member **226** must be such that it extends from the surface **224** to the bottom surface of the mold compound (i.e., the bottom surface of the semiconductor package). The combined thickness of the die pad extension member **226** and the die pad **220** is approximately equal to the thickness of the first horizontal member **230**, as shown. The die pad extension member **226** may be part of the die pad **220** (i.e., the die pad **220** and the die pad extension member **226** are a monolithic unit), or the die pad extension member **226** may be coupled to the die pad **220** using any suitable adhesive.

[0032] FIGS. **2**C**1**, **2**C**2**, **2**C**3**, **2**C**4**, and **2**C**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples. More specifically, FIGS. **2**C**1**-**2**C**5** depict the structure of FIGS. **2**A**1**-**2**A**5** being coupled to the structure of FIGS. **2**B**1**-**2**B**5**. In examples, the non-device side **205** of the semiconductor die **202** is coupled to the surface **222** of the die pad **220** using a suitable adhesive, such as a suitable die attach material. As shown in FIGS. **2**C**1**-**2**C**5**, die attach material **240** couples the semiconductor die **202** to the surface **222** of the die pad **220**. The die attach material **240** (examples of which include, e.g., solder and silver epoxy) is thermally conductive so that it can facilitate the transmission of heat from the semiconductor die **202** to the die pad extension member **226** via the die pad **220**, thereby efficiently removing heat from the semiconductor die **202** and from the semiconductor package more generally. The metal pillars **208** extending from the semiconductor die **202** are coupled to the routing structure **200** by way of solder bumps **210**. Any suitable number of metal pillars **208** may be included.

[0033] The method **100** also includes coupling a conductive terminal to the routing structure (**106**). As FIGS. **2**C**1**-**2**C**5** show, an adhesive, such as solder **242**, couples the conductive terminals **228** (e.g., the second horizontal members **232**) to the routing structure **200** (e.g., to one or more of the metal layers **204**).

[0034] As shown, the second horizontal member **232** of the conductive terminal **228** is closer to the routing structure **200** than is the surface **222** of the die pad **220** to which the semiconductor die **202** is coupled

[0035] In examples, the semiconductor die **202** has a thickness ranging between 50 microns and 250 microns, and in those specific examples, it is critical that the thickness of the die pad **220** range between 50 microns and 500 microns. In these specific examples, a thickness of the die pad **220** above this range is disadvantageous because it introduces an unacceptable degree of mechanical stress to the die **202** and unacceptably increase material costs, and a thickness of the die pad **220** below this range is disadvantageous because the die pad **220** will not adequately spread heat.

[0036] The method **100** further includes covering the semiconductor die, the die pad, the metal pillar, the routing structure, and the conductive terminal with a mold compound, the conductive terminal exposed to an exterior surface of the mold compound (**108**), and trimming the conductive terminal and the mold compound to form a semiconductor package (**110**). FIGS. **2**D**1**, **2**D**2**, **2**D**3**, **2**D**4**, and **2**D**5** are profile, alternative profile, top-down, bottom-up, and perspective views of a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples. More specifically, FIGS. **2**D**1**-**2**D**5** show a mold compound **244** covering the structure shown in FIGS. **2**C**1**-**2**C**5**. The die pad extension member **226** is exposed to the bottom surface of the mold compound **244**, as are the first horizontal members **230** of the conductive terminals **228**. The first horizontal members **230** of the conductive terminals **228** may also be exposed to side surfaces **246** of the mold compound **244**, such as to facilitate solder wetting.

[0037] Although the structure of FIGS. **2**B**1**-**2**B**5** is depicted as being a standalone structure, in practice, multiple such structures may be coupled to each other in a lead frame format. Tie bars, dam bars, and other such structures may be included to provide mechanical support, physical connections, control of mold compound flow, etc. After the mold compound **244** is applied, various portions of the lead frame may be trimmed, e.g., the conductive terminals **228**, tie bars, dam bars, and/or a mold compound strip covering multiple instances of the structure shown in FIGS. **2**C**1**-**2**C**5**. Such trimming produces a standalone semiconductor package **248**, as FIGS. **2**D**1**-**2**D**5** show.

[0038] As described above, the thermally conductive die attach material **240** and the die pad extension member **226** transmit heat away from the semiconductor die **202** and out of the semiconductor package **248**. To further facilitate this heat dissipation, through-silicon vias may be formed inside the semiconductor die **202**. FIGS. **2**E**1** and **2**E**2** are profile and perspective views of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples. More particularly, FIGS. **2**E**1** and **2**E**2** show through-silicon vias **250** extending through at least a portion of the thickness of the semiconductor die **202** and extending to the non-device side **205** of the semiconductor die **202**. By extending to the non-device side **205** and being exposed to an exterior of the semiconductor die **202** at the non-device side **205**, the through-silicon vias **250** make contact with the thermally conductive die attach material **240**, thereby efficiently collecting and transferring heat away from the circuitry of the semiconductor die **202**, through the die attach material **240**, through the die pad **220** and the die pad extension member **226**, and out of the semiconductor package **248**. In examples containing such through-silicon vias **250**, the distance between the circuitry formed in the device side **203** of the semiconductor die **202** and the ends of the through-silicon vias **250** most proximal to the device side **203** ranges between 5 microns and 20 microns, with a distance above this range being disadvantageous because of the lack of a robust heat dissipation pathway, and with a distance below this range being disadvantageous because it can result in brittleness of the die **202**. Further, although the vias **250** are referred to herein as through-silicon vias, the vias **250** also may extend through other types of semiconductor material.

[0039] Referring to FIGS. **2**A**1**-**2**F, in operation, signals generated by the semiconductor die **202** are carried to the metal layers **204** through the metal pillars **208**. The metal layers **204** and/or the vias **212** provide the signals to the second horizontal members **232**, which provide the signals to the first horizontal members **230** via the connecting members **234**. The first horizontal members **230** provide the signals to a PCB to which the semiconductor package **248** is coupled. Signals may be provided from the PCB to the semiconductor die **202** by the same electrical pathway.

[0040] It may be desirable to reduce the thickness of the semiconductor package **248** produced using the techniques described herein. It may also be desirable to position the semiconductor die **202** close to the die pad extension member **226**, and more specifically, closer to the bottom surface of the semiconductor package **248**, to improve heat dissipation from the semiconductor die **202**. Such improvements may be achieved in some examples by forming a cavity in the die pad **220** and positioning the semiconductor die **202** in the cavity. FIGS. **3**A**1**, **3**A**2**, **3**A**3**, **3**A**4**, and **3**A**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and a semiconductor die, in accordance with various examples. FIGS. **3**B**1**, **3**B**2**, **3**B**3**, **3**B**4**, and **3**B**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a die pad, die pad extension member, and conductive terminals, in accordance with various examples. FIGS. **3**C**1**, **3**C**2**, **3**C**3**, **3**C**4**, and **3**C**5** are profile, alternative profile, top-down, bottom-up, and perspective views, respectively, of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples. FIGS. **3**D**1**, **3**D**2**, **3**D**3**, **3**D**4**, and **3**D**5** are profile, alternative profile, top-down, bottom-up, and perspective views of a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples. Together, FIGS. **3**A**1**-**3**D**5** provide a process flow for manufacturing a semiconductor package **312** containing such a cavity and, thus, the improvements in package thickness and heat dissipation described above. The process flow of FIGS. **3**A**1**-**3**D**5** is similar or identical to the process flow of FIGS. **2**A**1**-**2**D**5**, with a few exceptions as follows. The structure shown in FIGS. **3**B**1**-**3**B**5** is identical to that shown in FIGS. **2**B**1**-**2**B**5**, except for the presence of a cavity **302** in the die pad **220**, and more specifically, in the surface **222** of the die pad **220**. The cavity **302** has a depth that ranges from 50 microns to 100 microns, with a depth greater than this range being disadvantageous because the remaining material thickness below the cavity **302** becomes unacceptably thin and fragile, and with a depth less than this range being disadvantageous because it will require the die to be unacceptably thin and thus vulnerable to damage. The cavity **302** has a floor **304**. Further, the structure shown in FIGS. **3**B**1**-**3**B**5** has conductive terminals **306**, each of which has a single axial center **308**, meaning that the conductive terminals **306** do not have separate segments that are in different vertical planes, as is the case in FIGS. **2**B**1**-**2**B**5**. Stated differently, each conductive terminal **306** has a uniform thickness and width along an entirety of a length of the conductive terminal **306**. FIGS. **3**A**1**-**3**A**5** are identical to FIGS. **2**A**1**-**2**A**5**, except for the presence of metal spacing members **310** on the routing structure **200**. When the structure of FIGS. **3**A**1**-**3**A**5** is coupled to the structure of FIGS. **3**B**1**-**3**B**5**, as shown in FIGS. **3**C**1**-**3**C**5**, the metal spacing members **310** provide adequate clearance for the semiconductor die **202**, the metal pillars **208**, and the solder bumps **210**. Stated another way, a combined thickness of the die pad extension member **226**, the die pad **220** below the cavity **302**, the semiconductor die **202**, the metal pillar **208**, and solder bump **210** between the metal pillar **208** and the routing structure **200** is approximately equal to a combined thickness of the conductive terminal **306** and one of the multiple metal spacing members **310**. The thickness of the metal spacing member **310** may be increased or decreased to enable appropriate contact between the solder bumps **210** and the routing structure **200**, as shown in FIG. **3**C**1**. The completed semiconductor package **312** is shown in FIGS. **3**D**1**-**3**D**5**.

[0041] The presence of the cavity **302** and the planar structure of the conductive terminals **306** decreases the total thickness of the semiconductor package **312**. Further, the positioning of the semiconductor die **202** closer to the die pad extension member **226** (due to the cavity **302**) increases heat dissipation. Further still, the presence of the cavity **302** reduces the weight of the semiconductor package **312** and the materials cost of the semiconductor package **312**.

[0042] FIGS. **3**E**1** and **3**E**2** are profile and perspective views of a routing structure and semiconductor die coupled to a die pad, a die pad extension member, and conductive terminals, in accordance with various examples. More specifically, FIGS. **3**E**1** and **3**E**2** depict that the semiconductor die **202** included in the semiconductor package **312** may include through-silicon vias **314** to enhance heat dissipation. The discussion of the through-silicon vias **250** provided above also applies to the through-silicon vias **314**, and thus the vias **314** are not described again here.

[0043] FIG. **3**F is a profile, cross-sectional view of the routing structure **200** in accordance with various examples. The routing structure **200** of FIG. **3**F is identical to that of FIG. **2**F, except that the routing structure **200** of FIG. **3**F includes metal spacing members **310**, as shown. The description provided above of the routing structure **200** in FIG. **2**F and the manufacture thereof also applies to the routing structure **200** shown in FIG. **3**F, and thus is not repeated here. The metal spacing members **310** may be applied by any suitable technique, such as a plating process (e.g., a plating process performed after the remainder of the routing structure **200** is manufactured).

[0044] FIG. **4** is a block diagram of an electronic device comprising a semiconductor package with enhanced heat dissipation and parasitic inductance mitigation, in accordance with various examples. Specifically, FIG. **4** shows an electronic device **400** comprising a printed circuit board

(PCB) **402**. A semiconductor package **404**, such as the semiconductor package **248** or the semiconductor package **312**, is coupled to the PCB **402**. A processor **406** and a transceiver **408** are coupled to the PCB **402**. The processor **406** is coupled to the semiconductor package **404** and the transceiver **408**. An antenna **410** is coupled to the transceiver **408**. The electronic device **400** thus may be an RF device capable of operating in any radio frequency band. The scope of this disclosure is not limited as such, and the electronic device **400** also may include non-RF devices.
[0045] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.
[0046] In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within +/-**10** percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

Claims

1. A semiconductor package, comprising: a semiconductor die having a device side including circuitry and a non-device side opposite the device side; a metal pillar coupled to the device side and extending away from the semiconductor die; a routing structure coupled to the metal pillar, the routing structure including: first and second metal layers, the first metal layer positioned closer to the device side than the second metal layer and coupled to the second metal layer by way of a via; and a dielectric covering the first and second metal layers and the via; a die pad coupled to the non-device side of the semiconductor die by way of a thermally conductive material; a die pad extension member coupled to an opposite surface of the die pad than the semiconductor die; a conductive terminal having a first horizontal member and a second horizontal member coupled to and vertically offset from the first horizontal member, the second horizontal member soldered to the first metal layer of the routing structure; and a mold compound covering the semiconductor die, the metal pillar, the routing structure, the die pad, and the conductive terminal, the first horizontal member and the die pad exposed to an exterior surface of the mold compound.
2. The package of claim 1, wherein the semiconductor die includes a through-silicon via in contact with the thermally conductive material and at least partially extending through a thickness of the semiconductor die.
3. The package of claim 1, wherein the thermally conductive material is solder or silver epoxy.
4. The package of claim 1, wherein the routing structure is not a printed circuit board (PCB).
5. The package of claim 1, wherein the dielectric includes AJINOMOTO® Build-Up Film (ABF).
6. The package of claim 1, wherein the second horizontal member of the conductive terminal is closer to the routing structure than is a surface of the die pad to which the semiconductor die is coupled.
7. The package of claim 1, wherein the semiconductor die has a thickness ranging between 50 microns and 250 microns and the die pad has a thickness ranging between 50 microns and 500 microns.
8. The package of claim 1, wherein a combined thickness of the die pad and the die pad extension member is approximately equivalent to a thickness of the first horizontal member.
9. A semiconductor device, comprising: a semiconductor die having a device side including circuitry and a non-device side opposite the device side; a metal pillar coupled to the device side and extending away from the semiconductor die; a routing structure coupled to the metal pillar, the routing structure including: first and second metal layers, the first metal layer positioned closer to

the device side than the second metal layer and coupled to the second metal layer by way of a via; and a dielectric covering the first and second metal layers and the via; a die pad having a cavity formed therein, the semiconductor die positioned at least partially inside the cavity, the non-device side of the semiconductor die coupled to a floor of the cavity by way of a thermally conductive material; a die pad extension member coupled to an opposite surface than the semiconductor die; a conductive terminal coupled to the first metal layer of the routing structure; and a mold compound covering the semiconductor die, the metal pillar, the routing structure, the die pad, and the conductive terminal, the conductive terminal and the die pad exposed to an exterior surface of the mold compound.
10. The device of claim 9, wherein the semiconductor die includes a through-silicon via in contact with the thermally conductive material and at least partially extending through a thickness of the semiconductor die.
11. The device of claim 9, wherein the thermally conductive material is solder or silver epoxy.
12. The device of claim 9, wherein the routing structure is not a printed circuit board (PCB).
13. The device of claim 9, wherein the dielectric includes AJINOMOTO® Build-Up Film (ABF).
14. The device of claim 9, wherein the semiconductor die has a thickness ranging between 50 microns and 250 microns and the die pad has a thickness ranging between 50 microns and 500 microns.
15. The device of claim 9, wherein the conductive terminal has a uniform thickness and width along an entirety of a length of the conductive terminal.
16. The device of claim 9, further comprising multiple metal spacing members on a surface of the routing structure facing the semiconductor die.
17. The device of claim 16, wherein a combined thickness of the die pad extension member, the die pad below the cavity, the semiconductor die, the metal pillar, and solder between the pillar and the routing structure is approximately equal to a combined thickness of the conductive terminal and one of the multiple metal spacing members.
18. The device of claim 16, wherein the multiple metal spacing members are coupled to the first metal layer of the routing structure.
19. The device of claim 9, wherein the device is a radio-frequency electronic device and the semiconductor die is a gallium nitride semiconductor die, and wherein the device includes a printed circuit board (PCB) to which the conductive terminal is coupled.
20. The device of claim 9, wherein the routing structure lacks air between the first and second metal layers.
21. A method for manufacturing a package, comprising: coupling a metal pillar extending from a device side of a semiconductor die to a routing structure, the device side having circuitry formed therein, the routing structure comprising: first and second metal layers, the first metal layer positioned closer to the device side than the second metal layer and coupled to the second metal layer by way of a via; and a dielectric covering the first and second metal layers and the via; coupling a non-device side of the semiconductor die opposing the device side of the semiconductor die to a die pad by a thermally conductive material; coupling a conductive terminal to the routing structure; and covering the semiconductor die, the die pad, the metal pillar, the routing structure, and the conductive terminal with a mold compound, the conductive terminal exposed to an exterior surface of the mold compound.
22. The method of claim 21, wherein the surface of the die pad is a floor of a cavity in the die pad.
23. The method of claim 21, wherein the semiconductor die has a thickness ranging from 50 microns and 250 microns.
24. The method of claim 21, wherein the semiconductor die includes a through-silicon via at least partially extending through a thickness of the semiconductor die.
25. The method of claim 21, further comprising a die pad extension member coupled to the die pad and extending away from the semiconductor die, wherein a combined thickness of the die pad and

the die pad extension member is approximately equivalent to a thickness of a segment of the conductive terminal.